Reproduction of BakhshiGanje et al. 2026, Phytobiomes J. 10:277−292

Fire blight canker microbiome − PacBio full−length 16S, BioProject PRJNA1348523 (48 samples: 32 canker, 16 healthy bark)

Re−analysis of public data on VT ARC TinkerCliffs − 2026−09−29

`Files: /projects/intro2gds/I2GDS2026/Team5/orlando/Claude_Code`

1. Summary

The public reads were re−analysed following the paper's Methods (QIIME2 2024.10, DADA2, SILVA 138, rarefaction, Bray−Curtis/PERMANOVA, Random Forest, LEfSe, Dirichlet−multinomial mixtures, FAPROTAX via microeco, SparCC networks). All computation ran on TinkerCliffs compute nodes (Slurm).

Reproduced well: read retention and ASV counts (1,839,176 reads / 2,333 ASVs vs 1,817,968 / 2,322), the canker community (top−5 genera Erwinia, Sphingomonas, Massilia, Pantoea, Novosphingobium in the paper's order), the canker−vs−healthy separation (PERMANOVA P = 0.001), LEfSe canker markers, DMM drivers (4 E. amylovora ASVs, Curtobacterium, Novosphingobium, Methylobacterium), the cultivar ranking of Erwinia ASVs (Cripps Pink > Gala > Aztec Fuji), FAPROTAX fermentation/plant−pathogen enrichment in cankers and methanol oxidation in healthy Gala > Aztec Fuji > unknown, and Erwinia network partners (+Pantoea, +Novosphingobium, −Sphingomonas roseiflava).

Main discrepancy: healthy bark reads are 99.8% apple chloroplast 16S (canker 53%); 8 of 16 healthy samples contain no bacterial reads after chloroplast removal. The paper does not mention chloroplast filtering and reports a Bacillus−dominated healthy microbiome; no Bacillus is found in healthy samples here. Consequently the healthy−bark composition, LEfSe healthy markers (Bacillus, Methylobacterium) and the healthy network (51 nodes in the paper, 2 here) cannot be reproduced from the public data.

Other differences: with chloroplast kept, cultivar is significant and the health x cultivar interaction is not (paper: the reverse); DMM model fit prefers k = 1 (paper presents k = 4); SparCC networks are much sparser than the paper's MENAP networks (canker T5: 251 vs 1,586 edges); FAPROTAX ureolysis does not differ (paper: the only N−cycle function that differs).

2. Pipeline (scripts in Claude_Code/scripts/, PDFs in Claude_Code/pdf/)

Step	What	Scripts	PDF
1	SRA/ENA metadata, sample sheet	01_get_metadata.sh, 01b_make_sample_sheet.py	–
2	Download 48 HiFi FASTQs (md5 checked)	02_download_reads.sh	–
3	FastQC + read length/quality summary	03_fastqc.sh, 03_read_summary.py	03_read_QC.pdf
4	QIIME2 import; DADA2 denoise−ccs (27F/1492R, 1,000−1,600 bp, maxEE 2, consensus chimeras)	04_dada2.sh, 04b_dada2_summaries.sh, 04_dada2_report.py	04_dada2_stats.pdf
5	SILVA 138 99% naive Bayes classifier (sklearn 1.4.2); chloroplast/mitochondria check	05_taxonomy.sh, 05_taxonomy_report.py, 05b_bacterial_read_check.py	05_taxonomy.pdf
6	Phylogeny (MAFFT/FastTree); rarefaction; alpha; Bray−Curtis PCoA; PERMANOVA (A: all, 14,881; B: canker bacteria, 505)	06_diversity.sh, 06_diversity_report.py	06_diversity.pdf
7	Composition, top−20 heatmap, Random Forest, LEfSe, DMM k = 1−7, Erwinia ASVs (bacteria−only, not rarefied)	07_composition_ml.sh / .R	07_composition_ml.pdf
8	FAPROTAX v1.2.6 via microeco (genus−level taxonomy)	08_faprotax.sh / .R	08_faprotax.pdf
8b	BLAST vs NCBI 16S RefSeq −> species; FAPROTAX again	08b_blast_species.sh, 08b_assign_species.py	08b_faprotax_species.pdf
9	SparCC (FastSpar, 1,000 bootstraps, |r| >= 0.2, P < 0.01); topology, Zi−Pi, Erwinia edges	09_networks.sh / .R	09_networks.pdf
10	This summary	10_summary_report.R	00_reproduction_summary.pdf

3. Paper vs this re−analysis (1/2)

Result	Paper	Ours	Verdict
Reads after chimera removal	1,817,968	1,839,176	match
ASVs	2,322	2,333	match
ASVs per canker / healthy sample	16−470 (mean 130) / 1−22 (mean 5)	19−471 (mean 138) / 1−12 (mean 4)	match
Canker top−5 genera (Fig. 4A)	Erwinia, Sphingomonas, Massilia, Pantoea, Novosphingobium	Same five, same order (46.6, 6.4, 4.1, 4.0, 4.0%)	match
Canker classes (Fig. 2A)	Gammaproteobacteria (up to 96%), Alphaproteobacteria	Gammaproteobacteria 69%, Alphaproteobacteria 17%	match
Healthy composition (Fig. 2A)	Bacilli, Acidobacteria	99.8% chloroplast; bacteria: Alphaproteobacteria, Gammaproteobacteria, Bacteroidia, Acidobacteriae; no Bacilli	NOT reproduced
PERMANOVA health status	R2 17.2%, F 9.35, P 0.001	A: R2 21.5%, F 16.3, P 0.001	match
PERMANOVA cultivar	n.s. (P 0.223)	A: P 0.002 (chloroplast share varies by cultivar); B: P 0.47	differs (A) / match (B)
PERMANOVA health x cultivar	R2 12.4%, P 0.016	A: R2 8.3%, P 0.263	differs
PCoA axes (Fig. 3)	23.7% / 6.9%	A: 62.7% / 10.3%	differs
Canker Simpson / Shannon (Fig. 2B)	~0.85−0.95 / ~1.5−5	B: 0.71−0.99 (mean 0.89) / 1.5−4.8 (ln)	match (B)

3. Paper vs this re−analysis (2/2)

Result	Paper	Ours	Verdict
Random Forest by cultivar (Fig. 4B)	Friedmanniella, Rhizobium grp, Cutibacterium, Nocardioides...	Rhizobium grp, Friedmanniella, Cutibacterium, Duganella... (OOB error 69%)	partly (weak model)
LEfSe canker (Fig. 5A)	Erwinia LDA > 5; Pantoea, Massilia, Kosakonia, Rhizobium grp...	Erwinia LDA 5.7; 36 canker markers, largely the same list (stable over 10 seeds)	match
LEfSe healthy (Fig. 5A)	Bacillus, Methylobacterium, Sphingomonas, Hymenobacter, Pseudomonas	Sphingomonas, Pseudomonas, Hymenobacter, Comamonadaceae; no Bacillus/Methylobacterium	partly
DMM clusters (Fig. 5B)	k = 4; sizes 4/5/11/12	k = 4 sizes 3/4/9/16; best fit k = 1	partly
DMM drivers (Fig. 5D)	4 x E. amylovora, Curtobacterium, N. fluoreni, Methylobacterium	Same (ASV0001−0004 E. amylovora, Curtobacterium, Novosphingobium, Methylobacterium)	match
Erwinia ASV ranking (Fig. 5C)	Cripps Pink > Gala > Aztec Fuji > Granniwinkle > Hewe's Crab > unknown	Cripps Pink (10) > Gala (8) > Aztec Fuji (6) > others (4)	match (top 3)
Ea species identity	BLAST: E. amylovora	10/13 Erwinia ASVs E. amylovora (99.7−99.9%, margin >= 0.9)	match
FAPROTAX (Fig. 6)	Canker: fermentation, plant pathogen; healthy: methanol oxidation (Gala > Fuji > unknown); ureolysis differs	Fermentation 56% vs 0; plant pathogen 47% vs 0 (with species); methanol oxidation Gala 62 > Fuji 25 > unknown 18; ureolysis n.s.	mostly
Canker network T5 (Fig. 7A, Table 1)	160 nodes, 1,586 links; hub Mucilaginibacter; avgCC 0.267	138 nodes, 251 links; max degree Mucilaginibacter; avgCC 0.267	partly (sparser)
Canker network T0 (Fig. 7C)	17 nodes; hub E. amylovora ASV1032	6 nodes; hub = 4 E. amylovora ASVs	partly
Erwinia edges (Fig. 7B/D)	+Pantoea, +Novosphingobium, −S. roseiflava ...	+Pantoea (0.50), +Novosphingobium (0.45), −S. roseiflava (−0.40)	match
Healthy network (Fig. 8)	51 nodes, 698 links, Bacillus hubs	2 nodes, 1 link (8 samples, 7−177 bacterial reads)	NOT reproduced

4. Decisions, deviations and caveats

DADA2 parameters are not given in the paper; standard PacBio CCS settings were used, with max length 1,600 bp (user's choice). Reads of ~1,800 bp (~19% of raw reads, 23% in healthy samples) are removed by this length filter; they are probably host mitochondrial rRNA but this was not tested.

Chloroplast: the paper's ASV counts and healthy alpha values match only with chloroplast kept, while its canker alpha values, class plots (no Cyanobacteria) and cultivar result match only with chloroplast removed. Step 6 therefore reports both (A: chloroplast kept, all samples; B: canker bacteria only). Steps 7−9 use the bacteria−only table.

Hypothesis (untested): the paper's healthy−bark 'Bacillus' may be mis−labelled chloroplast; our 14 chloroplast ASVs are assigned with confidence > 0.9997.

Species: SILVA 138 assigns no species; species were added by BLAST vs NCBI 16S RefSeq only when unambiguous (>= 99.5% identity, same genus as SILVA, >= 0.2 points ahead of the next species on full−length hits). E. pyrifoliae reaches 99.5% only on partial reference sequences.

Networks: MENAP could not be run; FastSpar (SparCC) with 1,000 bootstraps was used. 'Majority selection' was implemented as prevalence: T5 = ASV in >= 5 of 32 cankers, T0 = in more than half of cankers (as the paper defines 0), healthy T1 = in >= 1 healthy sample.

ASV IDs (ASV0001...) are ours, ranked by abundance; the paper's numbering (e.g. ASV1032) cannot be recovered. Lookup: results/07_composition_ml/asv_lookup.tsv.

Randomness: rarefaction (QIIME2, no seed option) may shift alpha values slightly between runs; R steps use fixed seeds (2026).

Failed/superseded runs are kept in logs/ with _FAILED_ or _superseded_ in the name (read−only home in container, metadata header, LEfSe empty result, marker_table class, broken pipe, species margin rule, bind_rows).

5. Where everything is

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Project folder: /projects/intro2gds/I2GDS2026/Team5/orlando/Claude_Code
  raw_data/     48 FASTQs (raw_data/fastq/), SraRunInfo.csv, ena_filereport.tsv, md5_check.txt, sample_metadata.tsv
  scripts/      every script, numbered by step (01–10)
  results/      QIIME2 artifacts (.qza/.qzv) and tables, one folder per step (03_fastqc ... 09_networks)
  pdf/          00_reproduction_summary.pdf (this report) and one PDF per step (03–09)
  reference/    SILVA 138 99% classifier (sklearn 1.4.2), NCBI 16S rRNA BLAST database
  logs/         Slurm logs of every job
  REPRODUCTION_SUMMARY.md   Markdown copy of this report
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Software: QIIME2 2024.10 (Apptainer, /apps/common/software/QIIME2/2024.10/); R 4.3.3 conda env ~/miniconda3/envs/canker_r (phyloseq 1.46, randomForest 4.7–1.2, DirichletMultinomial 1.44, microbiomeMarker 1.8, microeco 1.1, igraph 2.1.4); FastSpar env ~/miniconda3/envs/fastspar; BLAST+ 2.16.0.

QIIME2 .qzv files can be viewed at https://view.qiime2.org.